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Numerical Simulation of Multi-Fluid Shock-Turbulence Interaction

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Abstract. Accurate numerical simulation of multi-fluid Shock-Turbulence Interaction (STI) is conducted by a hybrid monotonicity preserving-compact finite difference scheme for a detailed study of STI in variable density flows. Theoretical and numerical assessments of data confirm that all turbulence scales as well as the STI are well captured by the computational method. Comparison of multi-fluid and single-fluid data indicates that the turbulent kinetic energy is amplified more and the scalar mixing is enhanced more by the shock in flows involving two different fluids/densities when compared with those observed in single-fluid flows.

INTRODUCTION

The interaction between a normal shock wave and an isotropic turbulence is an important fundamental problem that has been extensively studied. Understanding this problem is important to the modeling of Shock-Turbulence Interaction (STI) in various flow configurations. Early study of small amplitude turbulent fluctuations conducted by Kovasznay [1] showed the existence of three different modes in compressible turbulence: vorticity, acoustic and entropy modes. Ribner [2] proposed a theoretical model for the description of STI, in which Eulerian Rankine-Hugoniot equations were solved. In this theory, referred to as Linear Interaction Approximation (LIA), the pre-shock turbulence was assumed to be made of the above mentioned small amplitude disturbances, so nonlinear and viscous effects were excluded from the analysis and the changes of different modes across the shock were allowed to be investigated. The amplification of turbulent kinetic energy and the reduction of turbulent length scales by the shock, as predicted by LIA, were compared with the experimental and numerical results, including those generated by the “Direct Numerical Simulation” (DNS) method. Early DNS study of shock-isotropic turbulence by Lee et al. [3] was limited to relatively low flow Mach numbers (1.2 or less), and showed reasonably good agreement with LIA when turbulent Mach number M_t was kept small. Jamme et al. [4] and Mahesh et al. [5] have also used DNS to study the effects of different flow parameters in upstream turbulence on STI. Lee et al. [6], Larsson & Lele [7], Larsson et al. [8] have conducted DNS for different flow Mach numbers using a high order shock-capturing scheme, and found good agreement with LIA results for some of the statistics, like turbulent kinetic energy and vorticity variance at relatively low M_t . More recently, Ryu & Livescu [9] have conducted “shock resolving DNS” for a wide range of parameters and have showed that DNS results converge to LIA when the ratio of shock width to Kolmogorov length scale becomes small, even at low upstream Reynolds numbers.

All of the above studies (and most of other relevant works not discussed here) were focused on STI in single-fluid flows. The current work focuses on multi-fluid STI. Li and Jaber [10] have recently developed new shock-capturing finite difference methods based on a Monotonicity-Preserving (MP) scheme and have tested the method for different supersonic flows. This work is based on their method, extended to multi-fluids. Fifth-order MP scheme is used for shock capturing and sixth-order compact scheme with spectral-like resolution is used to resolve the turbulence. Details of the simulations and computational setup are presented in the following section. To show the accuracy of the simulations, results from convergence tests are then discussed. The following section starts with the

effects of density variations on STI and compares the multi-fluid results with those obtained from a reference single-fluid simulation. Analysis of multi-fluid STI data will then be presented before making the final conclusions.

MATHEMATICAL FORMULATION AND NUMERICAL METHOD

In the current study, a set of fully compressible, unfiltered Navier-Stokes equations is solved in their dimensionless form for a variable density (multi-fluid) Newtonian system. The dimensionless fluid viscosity is assumed to be dependent on the dimensionless temperature as $T^{0.75}$ and the ratio of the specific heats is 1.4. The Prandtl and Schmidt numbers are constant and equal to $Pr=Sc=0.75$.

In order to correctly resolve all turbulence time and length scales while keeping the results free of oscillations around the shock, a combination of a high-order compact scheme and the Monotonicity Preserving scheme is used in this study: the inviscid fluxes are computed using the 5th-order MP scheme, and the viscous fluxes are calculated using the 6th-order compact scheme. Time advancement is achieved using a third-order Runge-Kutta scheme. More details on numerical method and its implementation can be found in Li and Jaber [10].

COMPUTATIONAL SETUP AND NUMERICAL ACCURACY

The physical domain for the simulations is a box that has a dimension of 4π in the streamwise direction and $(2\pi, 2\pi)$ in the transverse directions. The coordinate system is fixed with respect to the initial normal shock position, so that during the simulation, the averaged shock position remains close to 2π in the streamwise direction. To prevent reflecting waves, a sponge layer is used at the end of the computational domain, where the turbulence is smoothly dissipated and the flow becomes laminar. Constant backpressure calculated from the Rankine-Hugoniot relation is used as the subsonic outflow condition. By using this outflow condition, the averaged shock position may be pushed upstream. However, due to the low turbulent Mach number, the shock speed is small enough that it does not affect the accuracy of turbulent statistics. In the transverse directions, the turbulence is assumed to be periodic and homogeneous. The flow is initially laminar and a normal planar shock wave is created in the middle of computational domain perpendicular to the flow by using the Rankine-Hugoniot relations. At the inflow, an isotropic turbulence is superposed on the uniform mean flow. The inflow turbulent fields are generated by conducting a separate simulation of decaying isotropic turbulence. The variable density (multi-fluid) effects arise from compositional variations, by correlating the density to an isotropic scalar field with a given RMS value. The initial spectrum of the scalar field is Von Karman with a peak at $k_0=8$. The ratio of the molar masses of the two fluids is 1.78, resulting in an Atwood number of 0.28. This value of the Atwood number was chosen such that the variable density effects are not negligible, yet the interaction with the shock wave is still in the wrinkled-shock regime. The mean flow Mach number is 2.0 and the turbulent Mach number is 0.09. The Reynolds number based on the Taylor micro length scale is 23. The turbulence is convected into the domain using Taylor's hypothesis, which is a good approximation since the turbulent Mach number is very low.

To resolve the turbulent flow length scales, the maximum turbulent wavenumber k_{max} , multiplied by the Kolmogorov length scale η is usually taken to be greater than 1.5. For the coarsest mesh used in our simulations, $k_{max}\eta$ equals to 3.21, proving that turbulent scales are well resolved. Furthermore, to correctly resolve the interaction between shock wave and turbulence, the scale separation between turbulent lengths and (numerical) shock thickness needs to be satisfied. As suggested by Ryu and Livescu [9], we consider the ratio of Kolmogorov length scale to the shock thickness as an important parameter for assessing this separation. Our estimates for the preliminary simulations based on just the $k_{max}\eta$ criterion suggest the lowest value of this ratio to be around 1.0, which indicates not a good scale separation. Obviously, to satisfy the scale separation criterion, the numerical shock thickness needs to be smaller than that resulted from the usual $k_{max}\eta$ consideration. A series of tests using different grid sizes around the shock region were conducted and the numerical shock thickness was found to correlate linearly with the grid size. Therefore, by using an extremely fine mesh around the shock, we successfully raised the ratio of Kolmogorov length scale to the shock thickness to around 5.0.

In our grid convergence tests, the same number of 256 uniform grid points were used in the transverse directions, but different meshes were used in the streamwise direction. A range of grid sizes around and away from the shock wave was tested. A uniform grid was also used to assess the reliability of grid compression around the shock wave. Figure 1 shows the grid sizes used in different simulations. The streamwise coordinate is denoted by k_0x , where k_0 is the peak wavenumber of the turbulent kinetic energy spectrum. The shock is located at $k_0x=0$.

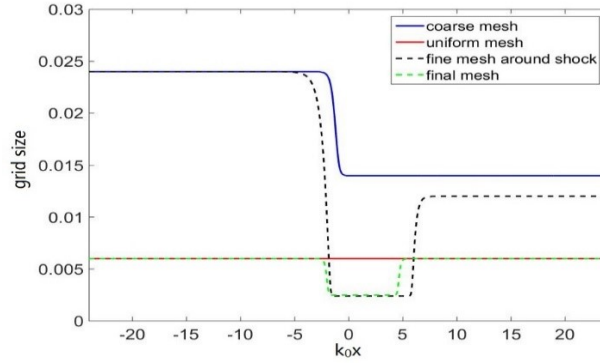


FIGURE 1. Grids used for convergence testing.

Two of the important flow/turbulence statistics obtained from the grid convergence tests, the turbulent kinetic energy and the Kolmogorov length scale, are shown in Figure 2. All the flow and scalar statistics are calculated by averaging in time and over homogeneous directions after reaching a statistically steady-state condition. The results in Figure 2 indicate the suitability of the grids for capturing the main flow features at large and small scales. The transverse turbulent kinetic energy spectrums (not shown) are also similar for different grids at all wavenumbers in both pre- and post-shock regions. Several other turbulent statistics are also compared. They all reflect similarly good grid convergence.

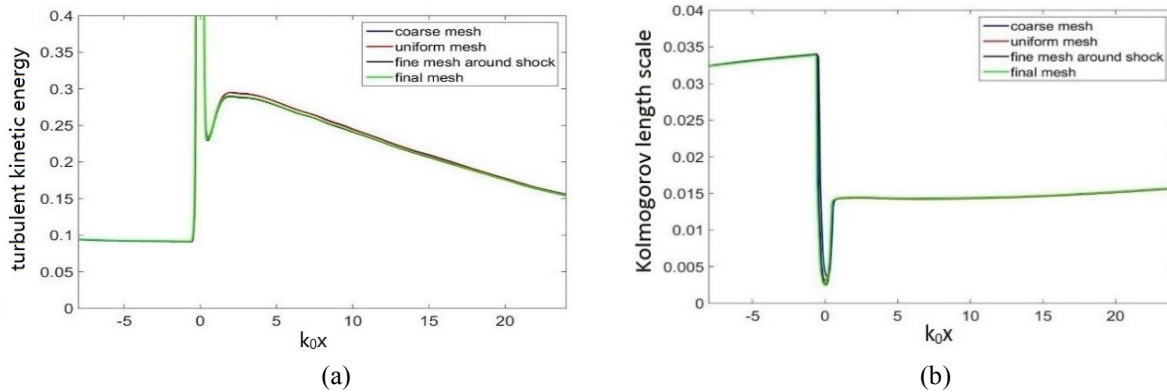


FIGURE 2. Results of grid convergence tests. (a) Turbulent kinetic energy, (b) Kolmogorov length scale.

RESULTS AND DISCUSSIONS

Two sets of statistics are presented in this paper; those obtained by Reynolds averaging are denoted by over bar ($\bar{\cdot}$) and those obtained by Favre-averaging are represented by over tilde ($\tilde{\cdot}$). The turbulent fluctuating parts are denoted by prime and double prime, correspondingly. In both averaging methods, the data collection starts after the flow reaches a statistically steady-state condition and continues for about two flow-through times. As explained in the previous section, the statistics are calculated by spatial and temporal averaging.

Comparison with Single-Fluid Flows

To study the multi-fluid or density variation effects on the shock-isotropic turbulence interactions, the multi-fluid results are compared with those obtained from a reference single-fluid simulation. The single-fluid simulation was conducted using the same inflow conditions for all variables but the density. A passive scalar equation is still solved to study the mixing.

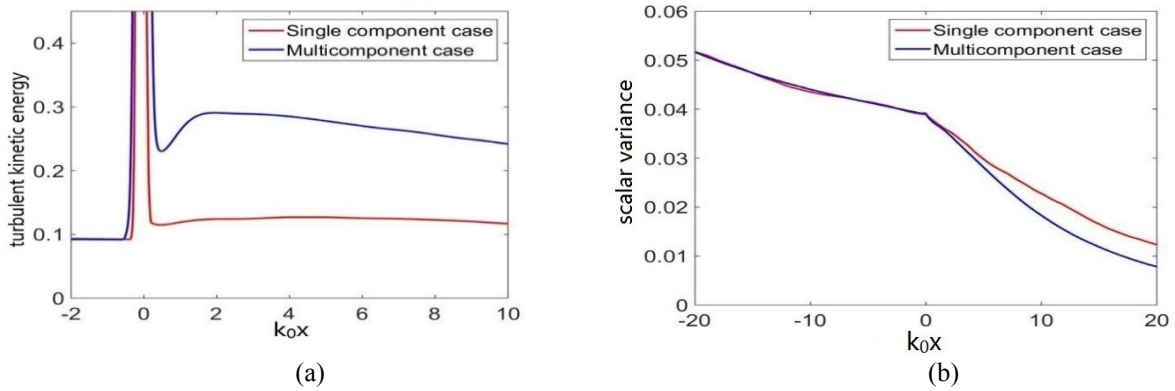


FIGURE 3. Comparison of statistics obtained from multi-fluid and single-fluid simulations. (a) Turbulent kinetic energy, (b) scalar variance.

Figure 3 compares two important flow statistics obtained from the multi-fluid and single-fluid simulations. Since these statistics are obtained by averaging over space and time, they are only dependent on the streamwise flow direction. However, due to wrinkling in shock surface and some unsteady shock movement, the averaged shock region is much larger than the local shock thickness. The shock movement and wrinkling (and therefore the average length of shock region) is more significant in the multi-fluid case.

In the pre-shock region, the statistics computed from single- and multi-fluid data are similar, confirming that the flow conditions are indeed similar in these two simulations. On the other hand, the comparison also shows that the variable density effects do not significantly alter the turbulence decay, at the relatively low Atwood number considered here. In the post-shock region, flow statistics like turbulent kinetic energy (Figure 3(a)) and vorticity variance (not shown) are amplified and those like turbulent length scales are reduced. When comparing the multi-fluid and the single-fluid cases, we observe that the amplification (or reduction) in turbulent statistics is more significant in the multi-fluid simulations in comparison to those seen in the single-fluid simulation. Mixing effects are also studied by comparing the scalar variance in Figure 3(b). The results in this Figure show that the density variations in the multi-fluid simulations have little effect on the scalar mixing in the pre-shock region, where the Atwood number is low. After passing through the shock, the scalar variance decreases much faster in the multi-fluid case, which proves that the density variation causes the turbulent mixing to be significantly enhanced following the shock interaction. The enhancement is expected to depend on the Atwood number, scalar field structure and so on.

Multi-Fluid Flows

In this section, various statistics computed from multi-fluid data are presented and discussed. In order to quantify the anisotropy generated by the shock, the ratios of the streamwise to spanwise statistics are calculated and plotted. Since the turbulence is isotropic in the pre-shock region, the streamwise values and spanwise values are very close in this region. After passing the shock wave, however, the turbulence becomes highly anisotropic due to the strong compression effect of the Mach 2.0 shock in the streamwise direction. Expectedly, the turbulence slowly returns to isotropy away from the shock. As shown in Figure 4(a), the ratios of vorticity variances, Reynolds stresses and Taylor microscales in the streamwise to spanwise directions are very close to 1.0 before the shock but they are far from unity and develop differently after the shock. For those statistics dependent more on smaller scales of turbulence like the vorticity variance, the return to isotropy is faster after the shock, consistent with the previous studies. The reduction in Taylor microscale across the shock (shown in Figure 4(a)) and the Kolmogorov length scale (shown before) suggest the enhancement of small-scale turbulence. The decrease in the scalar Taylor micro scale in Figure 4(b) also confirms that the small scale turbulence and mixing are significantly increased by the shock. Comparison between single- and multi-fluid results (not shown) indicates that this enhancement of small scales processes is much more significant in the multi-fluid case.

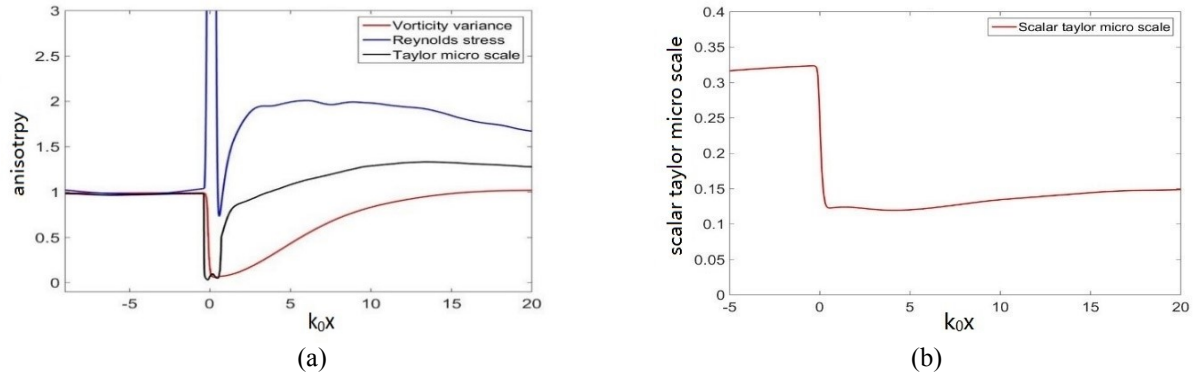


FIGURE 4. General statistics of multi-fluid case. (a) Anisotropy of Reynolds stress, Taylor micro scale and vorticity variance, (b) scalar Taylor micro length scale.

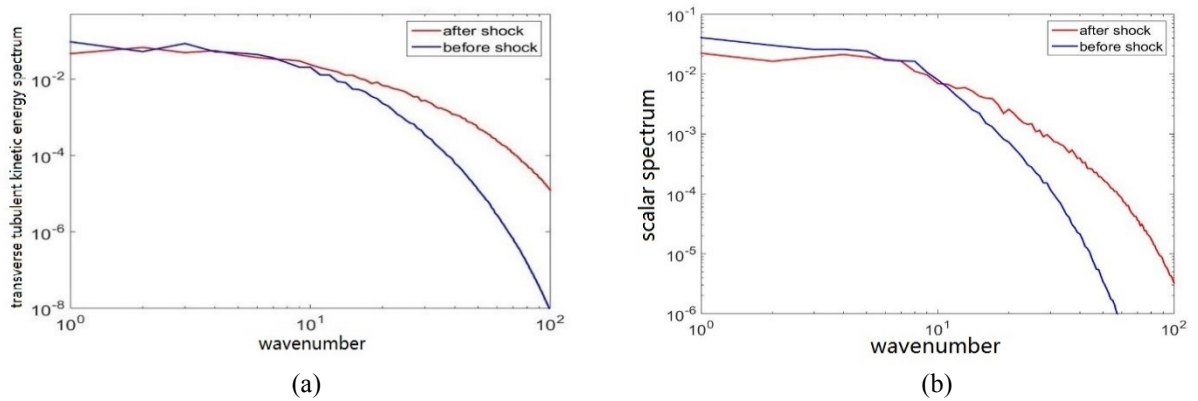


FIGURE 5. Statistics in wavenumber space. (a) Normalized transverse turbulent kinetic energy spectrum, (b) normalized scalar spectrum.

The effects of turbulent velocity and scalar interactions with the shock wave at different scales are further studied by comparing the normalized two-dimensional spectra of the instantaneous flow fields at a pre-shock location with those at a post-shock location situated near the peak of the streamwise velocity variance (Fig. 5). Evidently, there is considerable increase in energy and scalar variance at high wavenumbers. Again, this is much more significant in the multi-fluid case.

CONCLUSIONS

A hybrid numerical method, which combines a monotonicity-preserving scheme with a compact scheme, is used for high-order numerical simulations of an isotropic multi-fluid turbulence interacting with a planar shock wave. A reference simulation of single-fluid turbulence is also conducted with similar conditions. The goal is to study the effects of density variations (due to compositional changes) on the shock-turbulence interaction and mixing. Convergence tests are conducted to establish the accuracy of numerical results using different meshes with a wide range of grid sizes inside and outside the shock zone. The computed statistics are found to be independent of the grid when the turbulence after the shock is well resolved and the scale separation between numerical shock thickness and turbulent scales is adequate. Compared to the single-fluid reference case, turbulence amplification by the normal shock wave is much higher and the reduction in turbulent length scales is much more significant in the presence of density variations due to compositional changes. Turbulent mixing enhancement by the shock is also more important in the multi-fluid case. The enhancement is expected to depend on the Atwood Number, structure of the scalar field and so on. The post-shock development of the anisotropy indicates a faster return to isotropy at smaller turbulent velocity and scalar scales, consistent with previous studies.

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